

⊗ Cartesian Product :- $A \times B \Rightarrow x$ elements are in A and y elements are in B .

Set builder notation :- $A \times B = \{ (x, y) \mid x \in A \wedge y \in B \}$

Ex :- $A = \{1, 2, 3\}$
 $B = \{p, q\}$

$\therefore A \times B = \{(1, p), (1, q), (2, p), (2, q), (3, p), (3, q)\}$

$\swarrow \quad \searrow$
 $\in A \quad \in B$

Note :- $A \times B \neq B \times A$

Since $B \times A = \{(p, 1), (p, 2), (p, 3), (q, 1), (q, 2), (q, 3)\}$

Q :- when is $A \times B = B \times A$?

Ans :- when $A = B$, and when $A = \emptyset$ and $B = \emptyset$.

⊗ Example :- Prove that $(A \cup B) \times C = (A \times C) \cup (B \times C)$ using set builder

notations. Note x and y are variables :-

An :- take the arbitrary pair (x, y) in $(A \cup B) \times C$ where $x \in (A \cup B)$

and $y \in C$:-

$$\begin{aligned}
 (x, y) \in (A \cup B) \times C &= x \in (A \cup B) \wedge y \in C && \text{definition of } A \times B \\
 &= (x \in A \vee x \in B) \wedge y \in C && \text{definition of union } (\cup) \\
 &= y \in C \wedge (x \in A \vee x \in B) && \text{commutative} \\
 &= (y \in C \wedge x \in A) \vee (y \in C \wedge x \in B) && \text{distributive} \\
 &= (x \in A \wedge y \in C) \vee (x \in B \wedge y \in C) && \text{commutative} \\
 &= (x, y) \in (A \times C) \vee (x, y) \in (B \times C) && \text{definition of } A \times B \\
 &= (x, y) \in [(A \times C) \cup (B \times C)] && \text{definition of union } (\cup)
 \end{aligned}$$

⊗ Example :- Prove that

$$(A \cap B) \times (P \cap Q) = (A \times P) \cap (B \times Q)$$

using set builder notations. Note x and y are variables :-

An :- take the arbitrary pair (x, y) in $(A \cap B) \times (P \cap Q)$ where $x \in (A \cap B)$ and $y \in (P \cap Q)$:-

$$\begin{aligned}
 (x, y) &\in [(A \cap B) \times (P \cap Q)] \\
 &= x \in (A \cap B) \wedge y \in (P \cap Q) && \text{definition of } A \times B \\
 &= (x \in A \wedge x \in B) \wedge (y \in P \wedge y \in Q) && \text{definition of intersection } (\cap) \\
 &= (x \in A \wedge y \in P) \wedge (x \in B \wedge y \in Q) && \text{commutative and associative} \\
 &= (x, y) \in (A \times P) \wedge (x, y) \in (B \times Q) && \text{definition of } A \times B \\
 &= (x, y) \in [(A \times P) \cap (B \times Q)] && \text{definition of intersection } (\cap)
 \end{aligned}$$

⊗ Example :- Prove that $(A \times B) \setminus (C \times D) = (A \times (B \setminus D)) \cup ((A \setminus C) \times B)$

using set builder notations. Note x and y are variables :-

An :- take the arbitrary pair (x, y) in $(A \times B) \setminus (C \times D)$

$$\begin{aligned}
 (x, y) &\in (A \times B) \setminus (C \times D) \\
 &= (x, y) \in (A \times B) \wedge (x, y) \notin (C \times D) && \text{definition of subtraction} \\
 &= [(x, y) \in (A \times B)] \wedge \sim [(x, y) \in (C \times D)] && \text{definition of } \notin \\
 &= [x \in A \wedge y \in B] \wedge \sim [x \in C \wedge y \in D] && \text{definition of } A \times B \\
 &= [x \in A \wedge y \in B] \wedge [\sim x \in C \vee \sim y \in D] && \text{Demorgan (Logic)} \\
 &= [x \in A \wedge y \in B] \wedge [x \in \bar{C} \vee y \in \bar{D}] && \text{definition of complement} \\
 &= [x \in A \wedge y \in B] \wedge [x \notin C \vee y \notin D] && \text{definition of } \notin \\
 &= [x \in A \wedge y \in B \wedge x \notin C] \vee && \text{distributive} \\
 &\quad [x \in A \wedge y \in B \wedge y \notin D] \\
 &= [(x \in A \wedge x \notin C) \wedge y \in B] \vee && \text{associative and commutative} \\
 &\quad [x \in A \wedge (y \in B \wedge y \notin D)] \\
 &= [x \in (A \setminus C) \wedge y \in B] \vee && \text{definition of subtraction} \\
 &\quad [x \in A \wedge y \in (B \setminus D)] \\
 &= (x, y) \in [(A \setminus C) \times B] \vee && \text{definition of } A \times B \\
 &\quad (x, y) \in [A \times (B \setminus D)] \\
 &= (x, y) \in [(A \setminus C) \times B \cup A \times (B \setminus D)] && \text{definition of union } (\cup) \\
 &= (x, y) \in [A \times (B \setminus D) \cup (A \setminus C) \times B] && \text{commutative } \cup
 \end{aligned}$$